

Assignment no: 3

Results:

In the given assignment after performing necessary experiments we have observed the following results:

- **Classification report:**

	precision	recall	f1-score	support
0	0.87	0.95	0.91	5095
1	0.71	0.48	0.58	1422
accuracy			0.85	6517
macro avg	0.79	0.71	0.74	6517
weighted avg	0.83	0.85	0.83	6517

- **Confusion matrix:**

	Positive(predicted)	Negative(predicted)
Positive(predicted)	4820	275
Negative(predicted)	735	687

We have split the dataset with 80% in the train set and remaining 20% for test set. We have trained our decision tree model with train set and then evaluate it on test set. Our model performs with an accuracy of 0.8450207150529385

on test set.

Observations:

- In this dataset we have few columns that are originally in string format so we replaced them with corresponding integer values for model training and evaluation.
- There are few null and Nan (Not a number) entries so we have replaced such entries with mean (for null values) and median (for Nan) values of the corresponding attribute.

- Both the recall and f1score value for class 1 is 0.48 and 0.58 respectively which is very low in compare with its class 0 counterpart, which indicates even though the model correctly identifies 95% of all actual class 0 instances it only captures 48% of actual class 1 instances, meaning it misses more than half of them.
- The model performs very well on class 0 (majority class): high recall (0.95) and strong precision (0.87).
- The imbalance in support (5095 vs 1422) suggests the dataset is skewed, which often causes the model to favour the majority class. However, we have used log, square root and Yeo-Johnson transformation to remove skewness
- Overall, the classifier is biased toward class 0, achieving strong results there but struggling to detect class 1 reliably.